

Longest Match:

When a Packet arrive on the interface of a router the router takes several logical decisions based on the data plane to process the packet. Then the router compares the destination IP address of the packet with the IP routing table matching the longest-prefix route that match the destination IP address. The best route is the longest match route to the destination IP address. The route lookup process matches the destination IP address with the route available in the routing table and chooses the longest match route as forwarding route.

IP Packet Destination	172.16.0.10	10101100.00010000.00000000.00001010
Route 1	172.16.0.0/12	10101100.00010000.00000000.00000000
Route 2	172.16.0.0/18	10101100.00010000.00000000.00000000
Route 3	172.16.0.0/26	10101100.00010000.00000000.00000000

Longest Match to IP Packet Destination

